

Automated Construction of Multi-Element MEAM Interatomic Potentials

A DFT-Informed Neural-Network Optimization Framework

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- Elastic properties — the cubic stiffness constants C_{11} , C_{12} , and the derived Young's modulus E and Poisson's ratio ν — are the principal design criteria for structural alloys.
- Direct experimental characterization is costly and slow; classical molecular dynamics can recover these properties at negligible cost, *provided* an accurate interatomic potential is available.
- For metallic systems the **Modified Embedded Atom Method (MEAM)**¹ remains the de-facto standard semi-empirical formalism.

Objective

Construct MEAM potentials *automatically* for arbitrary subsets of elements, with no manual parameter fitting and no experimental input.

¹MEAM: Modified Embedded Atom Method.

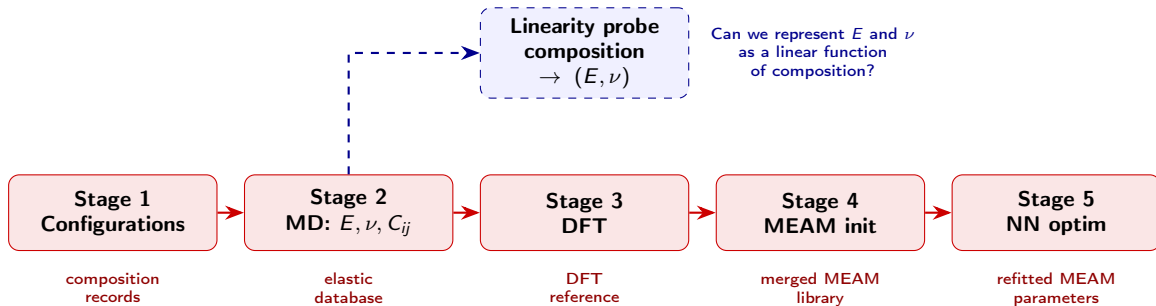


Published MEAM libraries cover disjoint subsets of elements; no single file spans the nine elements specified by the Al 7075 alloy.

Library File	Al	Co	Cr	Cu	Fe	Mg	Mn	Mo	Ni	Si	Ti	Zn
AlMgZn	✓					✓						✓
AlSiCuMgFe	✓			✓	✓	✓				✓		
CuMo				✓				✓				
FeMnNiTiCuCrCoAl	✓	✓	✓	✓	✓		✓		✓		✓	

- **Cross-interaction parameters** between elements drawn from distinct libraries are undefined; naive merging produces a potential that cannot be evaluated.
- **Purely data-driven interatomic potentials** (MLIPs)² avoid the coverage problem but require very large training corpora and forgo the physical interpretability of MEAM.

²MLIP: Machine-Learning Interatomic Potential.



MD: Molecular Dynamics; DFT: Density-Functional Theory; NN: Neural Network.



Stage	Description	Framework	Status
1	Random alloy configuration generation	Python	Complete
2	MD elastic-property evaluation	LAMMPS Python API	Complete
3	DFT reference calculations	Quantum Espresso + ASE	Complete
4	MEAM initialization & library merging	Python	Complete
5	Neural-network surrogate optimization	TensorFlow + LAMMPS	<i>Implementing</i>

Methodological Contribution

A closed-loop framework in which a neural-network surrogate of the MEAM-to-stiffness map is inverse-optimized against MD-measured targets, enabling end-to-end MEAM construction from first-principles input alone — no manual fitting, no experimental data.